Referee Report—"Leverage Risk and Investment: The Case of Gold Clauses in the 1930s"

This is a very nice paper studying the effects of leverage risk—the probability that leverage may increase for reasons out of a firm's control—on the firm's investment. Naturally, leverage is endogenous so it is usually difficult to find such situations, especially for developed countries with stable currencies. This paper studies the 1930s, when most US corporate debt was indexed to gold. Leaving the gold standard in 1933 implied a large devaluation of the dollar vis-à-vis gold, eventually of 67%, but lawsuits by creditors made this uncertain until the Supreme Court decided to allow such abrogation, initially passed by Congress in June 1933, in February of 1935. This paper finds that firms with a higher ratio of gold—indexed bonds to liabilities in 1932 saw a larger contraction in investment rates in 1933-34, and a larger increase in 35-36. These firms also paid out more (less) to equity holders over these periods, suggesting they were not experiencing financial constraints.

Comments and suggestions

I really like the paper. It presents a valuable contribution to the literature studying the effects of leverage risk, and capital structure more generally, as well as to the literature on the Great Depression, much of which has ignored firm level data. Moreover, the effect of the gold clauses has been important in the law and in the economics literature, but this is the first quantitative firm-level analysis on its effects on firms. And though currency or leverage risk may be relevant for firms in countries that do index their debts today, what makes the US in the 1930s quite unique is that it allows for studying the increase in uncertainty and its resolution before and after the Supreme Court decision. Thus, from an identification perspective, this setting has important advantages.

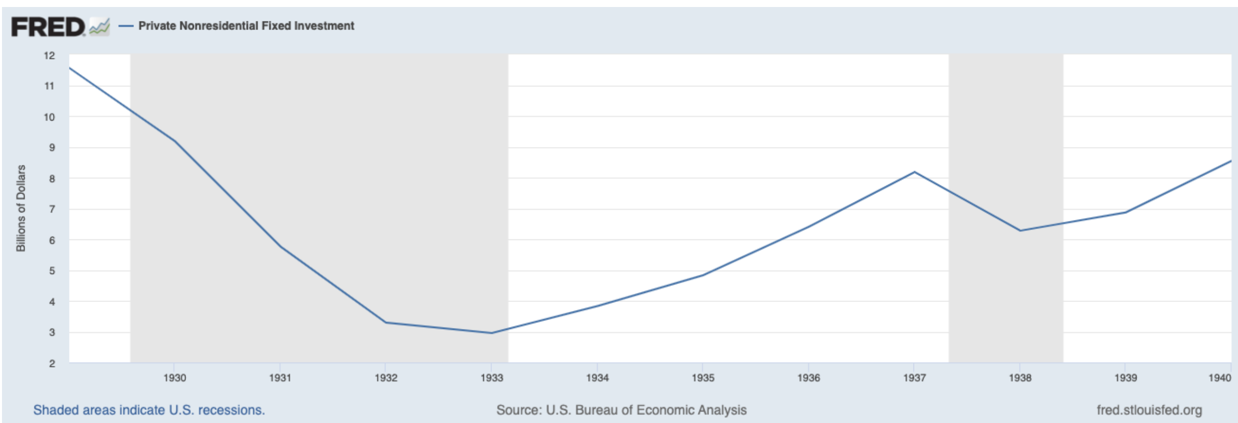
While I am enthusiastic about the paper, I do think that the empirical analysis and framing can be strengthened.

1. Gold clause or sensitivity to leverage

Ideally, to argue that gold-denominated debt is driving the results, one could want to contrast two firms with similar levels of <u>interest-bearing</u> debt but with variation in the fraction of that debt that is gold-denominated. The authors' main measure relates bonds outstanding with gold clauses relative to total liabilities, and controls for book and market leverage, measured by liabilities to assets. What this paper does not do, and what I think they may not be able to do, is to control for interest-bearing debt to total liabilities, or instead relate gold-denominated debt to interest-bearing debt as their main variable. The reason why I think they may not be able to do it is because the vast majority of bonds were indexed, and bank loans were tiny. What this means is that it is hard to know whether the indexing or the level of interest bearing debt are truly driving the results. One can think of reasons why variation in leverage may matter differentially over time independent of indexation. For example, bond markets were especially depressed in 1933 and 1934, but recovered substantially in 1935 and 1936. This is also when the second New Deal reactivated the economy, albeit temporarily. So it would be important to better disentangle gold-indexed bonds versus other bonds, as described above (and the summary statistics for these two should be listed in the paper). If this cannot be done, the paper needs to be more upfront about this important limitation.

2. Broader macro story, of just large firms?

Only the largest firms in the economy issued bonds. In fact, more than ¼ but less than ½ of the firms in the sample had gold-denominated bonds. Was this a more general issue in the economy at large, for firms or households? If yes, it would be great to provide some descriptive evidence. If not, it shapes how we see the contribution of the paper to our understanding of the Depression. Yes, understanding large firm behavior is important, but the macro estimates in the latest section of the paper could not be extrapolated to the rest of the economy, making the effects of this channel on the very sizable contraction in private non-residential fixed investment much smaller.



As shown above, most of the contraction in private fixed investment took place before 1932. Again, though the allocation within firms is significant, stating that "Taken together, our partial equilibrium aggregation exercise suggests that the uncertainty surrounding the abrogation of gold clauses led to a steep decline in public firms' investment and, therefore, likely contributed to the slow speed of economic recovery from the Great Depression. We find that leverage risk led public firms to delay investment in fixed capital for two years. This finding complements existing explanations of the delayed recovery based on the disruption of credit intermediation, which is more applicable to smaller firms. " seems much too strong, and the last sentence goes against the findings of Benmelech et al (2019), who show that the disruption in intermediation was quantitatively important for large firms early in the crisis. The statement that "large firms, with their ample cash holdings, were mostly insulated from the distress of the financial intermediation sector (see Lutz (1945), Hunter (1982), Bernanke (1983), and Calomiris and Mason (2003))" is also highly misleading because these papers are either not based on firm-level data or just have information on a handful of firms. Using over 1000 large firms, the cited paper by Benmelech clearly upends this view. This is acknowledged by Bernanke in his 2018 Brookings review article on the financial crisis. Firms needed cash for many reasons, not just investment, so the authors need to be much more cautious in their descriptions on constraints throughout.

3. Adjustments in di

I understand why the authors would want to keep di constant at the pre-abrogation decision level. But this does not mean that firms should not have strategically adjusted before, if they could. As the authors noted, in September 1931 UK left the gold standard, which was once unthinkable. This triggered a speculative attack on the dollar, under the belief that a devaluation in the US was imminent, and a banking panic in the US: https://www.federalreserve.gov/boarddocs/speeches/2004/200403022/default.htm. While the Fed chose to support the Gold Standard, it is quite reasonable to believe that uncertainty about a possible devaluation and what it would mean to borrowers should have been heightened. Especially since legal precedent was that contracts in gold would be upheld, one would expect unconstrained firms run by savvy managers to pay down their gold-denominated debts as much as possible. And to do so before December 1932. Thus, it is unclear one could treat the 1932 numbers as exogenous. Defining treatment as of 1932 is pre gold clause but not pre depression or pre dislocations in debt markets.

Similarly, it is unclear to me whether firms were able to pay interest and principal at the devalued or at the pre-abrogation levels from June 1932 to February 1935. If so, the reason why uncertainty may have affected activity is that they may have chosen to massively repay in case the Supreme Court decided against the devaluation. So they paid more to creditors and shareholders. Is that the case? Note that dropping firms with maturing bonds does not help here.

It would be nice to do validations with earlier ratios (1930 or ideally 1929), as well as to show the evolution of this ratio from 1929 to 1936.

One other historical validation would be to look at the downturn in 1937-38. By then gold denomination should not have mattered. Did interest bearing debt still affect investment choices then?

It would also be helpful to see what happened to the market value of firms by di around the September 1931 event.

If gold-denominated debt cannot be differentiated from bonds, then it is possible that the effects are instead driven by the constraints in making high interest payments during tough years (1933-34) relative to recovery years? Thus, there is a conflict of interest between creditors and shareholders, but it is not driven by the gold clause.

4. Differences in trends and panel structure

Another concern with defining treatment in 1932 is that more profitable firms in 1931-32 have less gold-denominated debt. It is possible that firms that were more recession proof paid their debt disproportionately, and also did better during tough years in 1933 and 1934. Ruling this omitted variable is challenging.

It is also hard within the structure of the data to argue that there are no pre-trends. I do not understand why the information is treated over two-year intervals, as opposed to using a panel structure. This would allow authors to show year-to-year estimates of the effects of di, ideally measured in 1930, over time, and show the specific years in which investment changes. It would also simplify the comparison over time. Similar magnitudes of the main coefficient over the two main periods with reversing signs are harder to contrast when the entire point is that investment declined over the first period.

It is also hard to understand the findings when the panel is not balanced, because firms may be dropping in and out of the sample for endogenous reasons. Say that only firms with di=0 go bankrupt. Would these have investment rate of minus infinity? A cleaner way would be to at least run a robustness check with a balanced panel.

5. Other comments

Why is investment rate trimmed and all other variables winsorized? Can you be consistent?

How can payout be negative for some firms?

Why is payout defined over fixed assets? This is certainly not standard.

Why is the sample restricted to publicly traded firms? Prices are almost not necessary, other than for the market leverage control, and this comes at the cost of losing about half the firms in the Moodys manuals.

Since tables report t-stats, it is harder to estimate the range of estimates. At least in the text, the paper should describe the confidence interval for investment effects.

It is not obvious that the effects of di should be linear. Especially since less than half of the sample has positive numbers, it would be good to assess where the effects are coming from.

Are the effects equally salient for small or large firms? Debt overhang may be worse for firms with fewer alternative sources of financing. If so, how does this affect the aggregate effect calculation?

The paper states that low ratings have no effect on investment, but it actually seems that the coefficient is too noisily estimated to be conclusive one way or the other. This is important because the paper seems to lever the difference in the ratings effect for investment and payouts.

In Table 2 it would be nice to add the number of firms for each column. In fact, I could not find a description of the number of firms with non-zero di.

Table 10 states that in Column (3) firms in regulated industries (transportation, communication, and utilities) are excluded. Surely it is strange that these firms were included in the industrials manual in the first place, so they should be dropped everywhere. In fact, these cannot be representative of the industries, which were covered in other manuals.